Machine Learning Libraries that can Make Web Applications Smarter

Web applications have come a long way. From the simple data capturing applications of the past they are now complex and dynamic. To enrich their capabilities further, it would be great to harness their machine learning potential. This article explores four Web browser based machine learning libraries and their features.

Web applications enable organisations to cater to the needs of a large number of users, distributed across various geographical locations, without requiring any local installation process. Upgrading Web applications is also comparatively simple, as the code needs to be updated only in the server. It is very tedious to update a desktop application in each installed location. Due to the advantages of scale and ease of maintenance, Web applications are now preferred to their desktop counterparts.

In the past, Web applications were comparatively simpler in nature. They mostly functioned as data collection platforms with simple interfaces. With the prolific growth in Web technologies, these apps have evolved into complex and dynamic entities.

Machine learning (ML) is evolving rapidly and is being applied to various domains. Web apps, too, can be enriched with ML capabilities and become more powerful. Machine learning can be incorporated into Web applications in two ways:

- Machine learning as a server side component
- Machine learning as a client side component

There are pros and cons for both these approaches. The server side applications have an advantage of better processing capabilities with a bigger memory. At the same time, one of the bottlenecks in server side ML is the delay due to network traffic. Each request with ML functionality needs to be communicated to the server, where it has to be processed, and the results should be returned to the client. These steps introduce latency in the application, which is best avoided.

To overcome this problem, it would be better to run the ML components in the browser itself. Thereby, the latency due to the network round-trip can be avoided. As the client devices these days have better processing capabilities, the ML components can be executed with an acceptable performance. This article explores four browser based ML libraries (Figure 1).

TensorFlow.js

TensorFlow is a popular machine learning library from Google. It makes the ML implementation effective. The TensorFlow.js is a JavaScript library, which can be used to implement ML models in Web browsers. Both training and deploying of the models can be done through TensorFlow.js, which has the following capabilities:

- The ML models can be built and trained from scratch with powerful APIs.
- It can be used to run the existing TensorFlow models inside the browser.
- The existing models can be retrained with TensorFlow.js.

The official documentation has provided many working examples (real-time human estimation, etc) of TensorFlow.js implementations that will give you an idea about the capabilities of this powerful ML library (*https://js.tensorflow.org/*).

If you are familiar with the working of Tensor, layers, optimisation and loss functions, then using the TensorFlow.js library is very simple.

TensorFlow.js can be included in Web applications in many ways. The simplest approach is to use a script tag, as